

Process Characterization Report

**A-Mab Drug Substance — Cation Exchange Chromatography (Step 7) —
PCR-007**

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AEX, anion exchange chromatography; ADCC, antibody dependent cell mediated cytotoxicity; AMV, analytical method validation; ANOVA, analysis of variance; CCD, central composite design; CEX, cation exchange chromatography; CPP, critical process parameter; CQA, critical quality attribute; Cpk, one sided process capability index; CV, coefficient of variation; DoE, design of experiments; DNA, deoxyribonucleic acid; DS, drug substance; ELISA, enzyme linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; IgG1, immunoglobulin G; KPP, key process parameter; LOQ, limit of quantitation; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; OD, optical density; PAR, proven acceptable range; PD, Process Development; PPQ, process performance qualification; qPCR, quantitative polymerase chain reaction; RMSE, root mean square error; RSM, response surface methodology; SD, standard deviation; SDM, scale-down model; SEC-HPLC, size exclusion high performance liquid chromatography; SOP, standard operating procedure; WC-CPP, well controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

Executive summary

Cation exchange chromatography is the polishing step of the A-Mab purification process and the last step in the train that changes the aggregate content of the product. It is operated in bind and elute mode on a strong cation exchange resin, in which the antibody binds below its isoelectric point and the column is washed at a controlled conductivity. The product is then eluted by a pH step and collected to a set optical density on the descending edge of the elution peak. Aggregate carries more positive charge than monomer and therefore binds more strongly and elutes late, which makes the aggregate content of the pool depend on how well the two species are resolved and on where the pool is stopped. Host cell protein, residual DNA and leached Protein A are cleared in the load flow through and in the wash. No viral clearance claim is made for this step.

This report presents the executed characterization of the step against the approved plan PCP-007. The studies were run on a qualified scale-down column model of the commercial process, which is operated at a nominal 15,000 L production scale. Of the 5 parameters carried into the study by RA-001, 4 were studied in a multivariate design and 1 was assessed one at a time. A two-level full factorial screening design of 19 runs identified the active factors, after which a face centred central composite design of 28 runs provided the predictive models for pool aggregate, pool host cell protein and step yield.

Protein load has the largest effect on all three responses. In addition, it governs host cell protein clearance together with the conductivity of the load and wash buffer, and aggregate clearance together with elution buffer pH and elution stop collect. The response-surface models explain 82.5 % to 97.5 % of the variation in their responses, and none of the three shows significant lack of fit against the centre-point pure error. Over the characterized region, 68.7% of the space meets both in-process criteria, and aggregate clearance is the binding constraint. Proven acceptable ranges were determined for each governed attribute against each response-surface factor and are reported in §7.

At commercial scale the step contributes to four drug substance attributes. Simulation of 2,000 batches with every parameter of the train inside its normal operating range gives a capability of 16.1 for aggregate and 6.1 for host cell protein, the tightest of the four attributes this step governs. The tightest capability of the drug substance as a whole is 1.51, which belongs to a viral clearance attribute that other steps set. Two deviations were recorded, 2 in total, and both were retained after investigation. The outcome of this step rolls up into the master report PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized immunoglobulin G monoclonal antibody whose primary mechanism of action is antibody dependent cell mediated cytotoxicity. It is produced by fed-batch culture and purified by the Novacyte platform train of Protein A capture, low-pH viral inactivation, cation exchange chromatography, anion exchange chromatography, small-virus retentive filtration and ultrafiltration with diafiltration. Cation exchange chromatography is Step 7 of that train, receiving the neutralized pool of the low-pH viral inactivation step and delivering the load for anion exchange chromatography.

The step is operated in bind and elute mode, in which the load and the wash are held at a defined conductivity and the product is eluted by a step change in buffer pH. The pool is collected from the ascending edge of the elution peak to a set optical density on the descending edge. Two separations happen in different phases of the cycle. Host cell protein, residual DNA and leached Protein A are acidic at the operating pH, do not bind the sulfonate ligand and leave in the load flow through and in the wash. Aggregate does bind, and more strongly than monomer, because a dimer presents twice the charged surface to the ligand, which is why it elutes behind the monomer. The aggregate the pool carries is set by the resolution between the two species and by the stop collect point.

The step forms no quality attribute of its own, but it clears four, and for one of them it is the last opportunity in the process. Because aggregate is formed upstream in the production bioreactor (Step 3) and is not reduced or generated by any step after cation exchange, the aggregate content of the pool this step delivers is the aggregate content of the drug substance. Host cell protein, residual DNA and leached Protein A are cleared again at anion exchange chromatography (PCR-008). Viral clearance is claimed for the low-pH inactivation, anion exchange and virus filtration steps, and no clearance is claimed here.

At the nominal set-point condition the step reduces pool aggregate from 1.91 % to 0.73 % of the total protein, a reduction of 2.6-fold, and reduces host cell protein from 9,125 ng/mg in the capture pool to 117 ng/mg, a reduction of 78-fold. Residual DNA is reduced by 1.5 log₁₀ and leached Protein A by 1.0 log₁₀, and the step yield is 95.2%.

1.2 Objectives

The study had five objectives.

- (i) Quantify the relationship between the process parameters of the step and pool aggregate, pool host cell protein and step yield, over ranges wider than the intended routine ranges.
- (ii) Define the multivariate region over which the step meets the in-process criteria for the attributes it governs.
- (iii) Establish a proven acceptable range for each parameter against each governed attribute, both at the set-point and with the remaining parameters varying within their normal operating ranges.
- (iv) Classify every parameter of the step against the criteria of SOP-4001, from the effects the study demonstrates.
- (v) Justify the operating ranges and the in-process controls that this step contributes to the control strategy for Stage 2.

1.3 Regulatory and scientific basis

The study is a Stage 1 process design activity in the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). It follows the enhanced development approach of ICH Q8, ICH Q9 and ICH Q11 (International Council for Harmonisation 2009, 2023b, 2012), in which prior knowledge and a risk assessment establish what needs to be studied, and designed experiments bound the region within which the process delivers acceptable quality. Because criticality is treated as a continuum and not as a binary label (International Council for Harmonisation 2023b), the attributes this step governs carry a graded criticality and the parameters a graded classification. The process model, the attribute framework and the risk methodology follow the published A-Mab case study (CMC Biotech Working Group 2009). Viral safety is evaluated under ICH Q5A (International Council for Harmonisation 2023a) for the three steps that carry a modular clearance claim (low-pH inactivation, anion exchange and virus filtration), of which this step is not one.

1.4 Related documents

Table 1.1 lists the documents this report depends on and the document it feeds. RA-001 set the scope of the study, PCP-007 defined the design and the acceptance criteria before execution, and PCMR-001 consolidates the outcome across the campaign.

Table 1.1: Related documents in the characterization package.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan

Docu- ment ID	Title	Relationship
PCP-007	Process Characterization Plan (Cation Exchange Chromatography (Step 7))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

Table 1.2 lists the controlled procedures under which the study was designed, executed and classified.

Table 1.2: Controlled procedures governing the study.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2010	Operation of the Cation-Exchange Polishing Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The cation exchange polishing step has been operated on the Novacyte humanized immunoglobulin platform for three previously licensed antibodies, X-Mab, Y-Mab and Z-Mab, and for the clinical and engineering supply of A-Mab (CMC Biotech Working Group 2009). The resin, the buffer system and the mode of operation are unchanged between those products and A-Mab. Because the platform experience establishes the mechanism of the step and the direction in which each parameter acts, the purpose of this study was to bound the ranges for A-Mab and not to discover the behaviour of the step.

Four mechanistic expectations follow from the physical chemistry of the separation and from that platform experience. Protein load sets the fraction of the ligand in use, and near capacity both the monomer and the aggregate peaks broaden and overlap, which was expected to reduce aggregate clearance and to reduce yield at a heavier load. Because the conductivity of the load and wash buffers sets the ionic strength that shields the charge on the ligand, a higher conductivity was expected to strip more weakly bound host cell protein during the wash, while the more strongly bound antibody remains adsorbed. The elution buffer pH sets the net charge on the antibody at the moment of elution. Therefore, a higher pH within the range was expected to weaken binding, to bring monomer and aggregate off closer together and to reduce the resolution on which aggregate clearance depends. The stop collect point decides how much of the descending edge enters the pool, and because the aggregate elutes on that edge it is the direct handle on the trade between aggregate clearance and yield.

Two attributes were expected to be insensitive to the operating parameters. DNA is strongly negatively charged and is repelled by the cation exchange ligand, which is why it passes through in the load and why its clearance is a property of the chemistry and not of any parameter setting. Leached Protein A and its fragments are acidic and do not bind under the load conditions, while the fraction complexed with the antibody is released when the complex binds the ligand. Consequently, the step delivers a modest and stable clearance. Both were measured across the study and neither was carried into the multivariate design.

2.2 Quality attributes in scope

Table 2.1 lists the drug substance attributes this step governs, with the acceptance criterion and the criticality assigned in RA-001. Criticality was assigned by the impact and uncertainty tool of the A-Mab case study (Tool #1), in which an attribute scored high or very high is called critical (CMC Biotech Working Group 2009). Aggregate is the attribute that drives the study, because it is scored high, its drug substance criterion is 5 % HMW, and this step is the last one that can reduce it. Host cell protein is scored moderate to high and is cleared further downstream at anion exchange chromatography. Residual DNA and leached Protein A are scored very low and low to moderate respectively, and both are cleared by a wide margin at this step and again at anion exchange.

Table 2.1: Drug substance quality attributes governed by the step.

CQA	Category	Acceptance	Criticality	Tool #1
Aggregates (HMW)	size variant	0-5 % HMW (SEC)	H	60
Host Cell Protein (HCP)	process impurity	0-100 ng/mg	M-H	36
Residual DNA	process impurity	0-0.001 ng/dose	VL	6
Leached Protein A	process impurity	0-5 ppm	L-M	16

The step is judged in this report against its own in-process criteria and not against the drug substance criteria in Table 2.1. An intermediate that carries 117 ng/mg host cell protein against a drug substance limit of 100 ng/mg is not a failing intermediate, because anion exchange chromatography reduces it a further 6.2-fold. Each in-process criterion is therefore derived from the drug substance criterion by carrying it back through the clearance the downstream train delivers and dividing by an assurance factor. For aggregate no step downstream reduces the attribute, which leaves the ceiling at the drug substance criterion itself. An assurance factor of 5 gives an in-process limit of 1 % HMW. For host cell protein the ceiling carries back through anion exchange (Step 8), where an assurance factor of 3.2 gives an in-process limit of 193 ng/mg. Both criteria sit well below the point at which the drug substance would be at risk, which is what makes them controls and not break-even points.

2.3 Risk-based prioritization and parameter selection

RA-001 ranked every operating parameter of the step by its potential impact on a quality attribute and by its potential to interact with other parameters, and assigned each parameter to a study type on the resulting score. Protein load, wash conductivity, elution buffer pH and elution stop collect were ranked highest and were carried into

the multivariate design, because each was expected to act on aggregate clearance or on host cell protein clearance and because the first three were expected to interact. Elution flow rate was ranked lower and was assigned to a univariate assessment, because within the range considered and at the particle size of the resin the effect on peak sharpness is small. Parameters held to platform specification (the buffer composition, the column packed bed height and the resin lot) were not carried into the study. They are identity and quality controlled under SOP-2010 and SOP-2008, and were not characterized in this study.

The ranges carried into the study were widened beyond the intended routine ranges by a factor of two to three on each parameter, in order to find an edge of failure where one exists inside a range the process could plausibly reach. The characterization ranges and the routine ranges are given in Table [4.1](#).

3 Materials and methods

3.1 Scale-down model and its qualification

The studies were executed on a laboratory-scale packed column model of the commercial cation exchange step, qualified under SOP-1001. The model uses the same resin, the same buffer system and the same packed bed height as the commercial column, at the same linear velocity and at the same load, wash and elution volumes expressed in column volumes. Column efficiency was matched by plate count and peak asymmetry before each campaign. The parameters that scale with column diameter, principally the volumetric flow rate and the absolute load mass, were set to preserve the commercial residence time and the commercial load in grams of protein per litre of resin.

Model qualification compared the scale-down column with at-scale data (engineering and clinical manufacture) at the target condition. The comparison covered the input attributes of the load, the output attributes of the pool and the performance of the separation itself, which are aggregate, host cell protein, residual DNA, leached Protein A, step yield and the elution peak volume. No difference was found between scales that the run to run spread of the at-scale data did not already contain. Therefore, the model is suitable for defining operating ranges that will be applied at commercial scale, and through that qualification the claims in this report are made for the commercial step.

The centre-point replicates of each design carry the pure error of the model, with 4 replicates in the response surface design and 3 in the screening design. Their spread is the reference against which the lack of fit of each fitted model is tested in §5.3, and measures the reproducibility of the scale-down system as a whole, including the analytical methods.

3.2 Operation

Each run followed the cycle defined in SOP-2010. The column was equilibrated in the load and wash buffer at the conductivity set for that run, loaded to the protein load set for that run, washed with the same buffer, and eluted by a step change to the elution buffer at the pH set for that run. The pool was collected from a fixed optical density on the ascending edge of the elution peak to the stop collect optical density set for that run on the descending edge. The column was then stripped, sanitized and stored under SOP-2008.

The load material for every run was the neutralized pool of the low-pH viral inactivation step, prepared from a single production campaign so that the input aggregate and host cell protein burden would not vary between runs. Resin lot, packed bed height, buffer composition, equilibration volume and wash volume were held at their platform values throughout. These inputs are controlled by identity and release testing under SOP-2010 and SOP-2008 and were not varied in the study.

3.3 Analytical methods

Table 3.1 lists the validated methods used to measure the responses and the cleared attributes. Pool aggregate was measured by size exclusion chromatography (SEC-HPLC) under AMV-3011 and is reported as the percentage of high molecular weight species in the total peak area. Pool host cell protein was measured by immunoassay (ELISA) under AMV-3012 and is reported relative to the product mass. Residual DNA was measured by quantitative polymerase chain reaction (qPCR) under AMV-3014 and leached Protein A by immunoassay under AMV-3016. Step yield was calculated from the product mass in the pool relative to the product mass loaded, each determined by ultraviolet absorbance.

Table 3.1: Validated analytical methods used in the study.

Reference	Title	Type
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation

The precision of these methods sets a floor on the variation the study can attribute to the process. The host cell protein immunoassay (AMV-3012) is the least precise of the four, at 9.5 % coefficient of variation. Its share of the observed variance in the pool host cell protein result is 0.40. The leached Protein A immunoassay has a precision of 6.5 % and a limit of quantitation of 0.25 ppm, and method performance is summarized in Appendix D.

3.4 Sampling plan

The load was sampled once for each run and assayed for aggregate and host cell protein, which made the burden entering the column known for every run. The elution pool was sampled after collection was complete and assayed for aggregate, host cell protein, residual DNA and leached Protein A. Protein concentration was measured on the load and on the pool for the yield calculation. Runs were executed in randomized order within each design.

The screening design contributed 19 runs and the response surface design 28 runs, of which 3 and 4 respectively were centre-point replicates at the set-point condition of every factor. Every run of both designs is reported, and no run was excluded from any analysis.

3.5 Statistical methods

Each factor was coded so that its set-point is 0 and the edges of its characterization range are the coded levels -1 and $+1$. The set-point is the midpoint of the characterization range for all four multivariate factors of this step, which makes the centre point of both designs (the coded origin) the set-point condition of the step. Analyses were run under SOP-1002 at a significance level of $\alpha = 0.05$.

The screening data were fitted by ordinary least squares to a model containing the four main effects and the six two-factor interactions. The effect of a term is reported as twice its coded coefficient, which is the change in the response across the full characterization range of that factor. The response surface data were fitted to the full quadratic model in the same four factors, which adds the four squared terms. Model adequacy was judged by the coefficient of determination, by its adjusted form, by the predicted form computed from the prediction sum of squares, by the overall F test, and by an analysis of variance that splits the residual into lack of fit and pure error and tests the first against the second. A factor with no term significant at $\alpha = 0.05$ in any model is reported as having no detected effect, which is a statement about the ranges studied and not about the parameter in general.

The screening design identifies which factors are active, while the response surface model is the predictive model, from which every prediction, design space boundary and proven acceptable range in this report is taken. The screening estimates are reported in §5.2 for the directions and the relative sizes they establish, and are not used to predict.

Proven acceptable ranges were determined in two analyses for each attribute and parameter. The first holds the other parameters at their set-points and scans the parameter of interest across its characterization range on the fitted response surface. The second varies the other parameters within their normal operating ranges by Monte-Carlo simulation of the fitted model, including the residual variation of the model, and requires the resulting 95 % predictive interval to remain inside the criterion. The second is the more demanding of the two, and is the analysis that bounds the routine operating region. Commercial-scale capability was estimated by simulating 2,000 batches of the whole train with every parameter varying inside its normal operating range, and is reported as a one sided capability index against the drug substance criterion.

4 Study design

4.1 Factors, ranges and the knowledge space

Table 4.1 lists every parameter of the step with its set-point, its normal operating range, the range studied and the classification the study led to. The characterization ranges bound the knowledge space of the step, within which protein load was studied from 10 to 40 g/L resin against a normal operating range of 18 to 32 g/L resin. The other three multivariate factors were widened in the same proportion. The load buffer and the wash buffer are the same buffer, whose conductivity is one parameter of the step and is called the wash conductivity in the text below. The classification column is the outcome of the study and is justified parameter by parameter in §9.

Table 4.1: Parameters of the step, ranges studied and final classification.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Protein load	g/L resin	25	18-32	10-40	WC- CPP	multivari- ate
Load/Wash conductivity	mS/cm	5	4-6	3-7	WC- CPP	multivari- ate
Elution buffer pH	pH	6	5.85-6.15	5.8-6.2	WC- CPP	multivari- ate
Elution stop collect	OD desc.	1	0.7-1.3	0.5-1.5	WC- CPP	multivari- ate
Elution flow rate	cm/hr	200	150-250	100-300	GPP	univariate

Table 4.2 gives the letter codes used for the factors in the effect and coefficient tables and in the design matrices of Appendix A and Appendix B.

Table 4.2: Coded factor legend.

Code	Factor	Unit	Studied in
A	Protein load	g/L resin	screening + RSM
B	Load/Wash conductivity	mS/cm	screening + RSM
C	Elution buffer pH	pH	screening + RSM
D	Elution stop collect	OD desc.	screening + RSM

4.2 Screening design

The screening study was a two-level full factorial in the four multivariate factors, augmented with 3 centre points, giving 16 factorial runs and 19 runs in total. The design estimates all four main effects and all six two-factor interactions, and leaves 8 residual degrees of freedom (Appendix A). A full factorial was used rather than a fraction because four factors are few enough that no aliasing between a main effect and a two-factor interaction is necessary, and because three of the four were expected to interact. The centre points give a direct test for overall curvature and the pure error term. The coded design matrix and the measured responses of the screening design are in Appendix A.

4.3 Response-surface design

The response surface study was a face centred central composite design in the same four factors. It reuses the 16 factorial points, adds 8 axial points at the face centres of the coded cube and 4 centre points, giving 28 runs. Axial points at the faces rather than beyond them keep every run inside the characterization range. No prediction from the model therefore requires a setting the study did not visit. The design supports the full quadratic model, which has 15 terms and leaves 13 residual degrees of freedom, of which 3 are pure error. The coded design matrix and the measured responses of the response surface design are in Appendix B.

4.4 Univariate assessment

Elution flow rate was assessed one at a time across 100 to 300 cm/hr with the other parameters at their set-points. Flow rate sets the residence time during elution and therefore the sharpness of the elution peak. However, over this range and at the particle size of the resin the change in resolution is small. The assessment found no effect on pool aggregate, pool host cell protein or step yield that would place the parameter in the design space. It is classified as a general process parameter in §9 and is controlled at its set-point by the pump under SOP-2010.

5 Results

5.1 Centre-point performance and reproducibility

Table 5.1 gives the mean, the standard deviation and the coefficient of variation of the 4 centre-point replicates of the response surface design. These runs were executed at the set-point of every factor and are the reproducibility of the scale-down system as a whole, including the analytical methods.

Table 5.1: Centre-point reproducibility of the response-surface design.

Response	n	Mean	SD	%CV
Aggregates (HMW, %)	4	0.771	0.043	5.58
Pool HCP (ng/mg)	4	134	18.7	13.9
Step yield	4	0.918	0.0289	3.15

Step yield is the most reproducible response, at 3.2 %, and pool aggregate follows at 5.6 %. Pool host cell protein is the least precise of the three, at 13.9 %, and most of that spread is analytical. The immunoassay used to measure it (AMV-3012) has a precision of 9.5 % on its own and accounts for 0.40 of the variance of the reported result. The process contribution to the centre-point spread of pool host cell protein is therefore smaller than the total. The screening design gives the same ordering on 3 replicates, with 1.4 % for step yield, 10.2 % for pool aggregate and 17.6 % for pool host cell protein.

Therefore, an effect on pool host cell protein must be larger than an effect on pool aggregate to be detected with the same confidence. The centre-point means also sit close to the values the nominal process delivers, at 0.77 % HMW against the nominal 0.73 % and 134 ng/mg against the nominal 117 ng/mg. This is one of the checks that the scale-down column reproduces the commercial step.

5.2 Screening: factor effects

Table 5.2 summarizes the four-factor screening models, all three of which are significant overall and carry a high coefficient of determination.

Table 5.2: Screening model summary for the three responses.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Aggregates (HMW, %)	19	0.973	0.939	0.869	28.5	3.57e-05	0.096
Pool HCP (ng/mg)	19	0.936	0.857	0.467	11.8	0.000937	35.4
Step yield	19	0.924	0.829	0.498	9.74	0.00181	0.0115

The gap between the coefficient of determination and its predicted form is the first indication of how much of each screening fit will survive into a predictive model. For pool aggregate the two are 0.973 and 0.869, for pool host cell protein 0.936 and 0.467, and for step yield 0.924 and 0.498. A model that explains most of the variation it was fitted to and predicts a held out run poorly is carrying terms it cannot support. Therefore, the screening design is read here for the directions and the relative sizes of the effects, while the response-surface models of §5.3 are used to predict.

Pool aggregate is affected by protein load, elution buffer pH and elution stop collect, and Table 5.3 gives the six largest effects. Protein load has the largest effect, at +0.60 % HMW across its characterization range. Elution buffer pH follows at +0.34 % and elution stop collect at +0.32 %, and all three are significant at $p < 0.001$. The three interactions between those same factors are significant, the largest being protein load with elution buffer pH at +0.20 %. Wash conductivity has no effect on pool aggregate, at -0.018 % ($p = 0.71$).

Table 5.3: Six largest screening effects on pool aggregate.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	0.599	0.299	0.024	12.5	1.6e-06	***
C	0.34	0.17	0.024	7.09	0.000103	***
D	0.319	0.16	0.024	6.65	0.000161	***
AC	0.196	0.0979	0.024	4.08	0.00354	**
AD	0.154	0.0772	0.024	3.22	0.0123	*
CD	0.131	0.0657	0.024	2.74	0.0256	*

Pool host cell protein is affected by wash conductivity and by protein load, the two interact, and Table 5.4 gives the six largest effects. Raising the wash conductivity across its range lowers pool host cell protein by 127 ng/mg, while raising the protein load across its range raises it by 116 ng/mg. The interaction between them is -57 ng/mg and is significant ($p = 0.012$). Neither elution buffer pH nor elution stop collect has a significant effect on pool host cell protein in the screening design.

Table 5.4: Six largest screening effects on pool host cell protein.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-127	-63.3	8.85	-7.15	9.7e-05	***

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	116	58.2	8.85	6.58	0.000173	***
AB	-56.7	-28.4	8.85	-3.21	0.0125	*
BD	39.6	19.8	8.85	2.24	0.0557	
D	-25.8	-12.9	8.85	-1.46	0.182	
AD	-24.6	-12.3	8.85	-1.39	0.202	

Protein load is the only parameter that affects step yield, which falls by 5.2% of the loaded product across the characterization range of the load, at $p < 0.001$. No other term reaches significance, the largest of them being the interaction of wash conductivity with elution buffer pH ($p = 0.07$). The full effect tables for all three responses are in Appendix C.

5.3 Response-surface models

Table 5.5 summarizes the three full quadratic models fitted to the response surface data.

Table 5.5: Response-surface model summary for the three responses.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Aggregates (HMW, %)	28	0.975	0.949	0.823	36.9	3.85e-08	0.0749
Pool HCP (ng/mg)	28	0.966	0.929	0.827	26.3	3.07e-07	16.8
Step yield	28	0.825	0.637	0.196	4.38	0.00573	0.0188

The models for pool aggregate and pool host cell protein are adequate for prediction, and each explains more than 97 % of the variation in its response. The adjusted and the predicted forms stay close to the coefficient of determination, and the predicted form is 0.82 for pool aggregate and 0.83 for pool host cell protein. The model for step yield is weaker, with a coefficient of determination of 0.83 and a predicted form that falls to 0.20, because only one term in the model is significant and the response varies little across the design. The yield model is used in this report to describe the direction and size of the load effect and is not used to predict a yield at a new condition.

Table 5.6 and Table 5.7 partition the residual of each governed response into lack of fit and pure error and test the first against the second, and neither test is significant. The lack of fit p value is 0.16 for pool aggregate, 0.68 for pool host cell protein and 0.96 for step yield. The quadratic model therefore describes the data to within the reproducibility of the scale-down system, and there is no evidence of a shape the model has missed.

Table 5.6: Analysis of variance and lack-of-fit test for pool aggregate.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	2.89	14	0.207	36.9	3.85e-08
Residual	0.0729	13	0.0056	nan	nan
Lack of fit	0.0673	10	0.00673	3.63	0.158
Pure error	0.00556	3	0.00185	nan	nan

Table 5.7: Analysis of variance and lack-of-fit test for pool host cell protein.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	104,028.0	14	7,430.6	26.3	3.07e-07
Residual	3,670.2	13	282.3	nan	nan
Lack of fit	2,626.7	10	262.7	0.755	0.679
Pure error	1,043.5	3	347.8	nan	nan

Table 5.8 gives the coefficients of the pool aggregate model, in which the three main effects that screening identified are confirmed, at +0.280 for protein load, +0.172 for elution buffer pH and +0.157 for elution stop collect per coded unit. The three interactions between them remain significant, and wash conductivity remains without effect ($p = 0.14$). No squared term is significant, the smallest p value among the four being 0.11, for protein load. The aggregate surface is therefore close to planar across the ranges studied, with its shape coming from the interactions and not from curvature.

Table 5.8: Response-surface coefficients for pool aggregate.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.751	0.0259	29	3.36e-13	***
A	0.28	0.0176	15.9	6.77e-10	***
B	0.0276	0.0176	1.57	0.142	
C	0.172	0.0176	9.75	2.41e-07	***
D	0.157	0.0176	8.88	7.05e-07	***
AB	0.00964	0.0187	0.515	0.615	
AC	0.096	0.0187	5.13	0.000193	***
AD	0.0851	0.0187	4.54	0.000551	***
BC	-0.00211	0.0187	-0.113	0.912	
BD	0.0137	0.0187	0.733	0.477	
CD	0.0694	0.0187	3.71	0.00262	**
A ²	0.0798	0.0466	1.71	0.111	
B ²	-0.00875	0.0466	-0.188	0.854	
C ²	0.0264	0.0466	0.566	0.581	
D ²	0.0677	0.0466	1.45	0.17	

Table 5.9 gives the coefficients of the pool host cell protein model. Protein load and wash conductivity have coefficients of nearly equal size and opposite sign, at +52.5 and -49.7 ng/mg per coded unit. Two interactions are significant, protein load with wash conductivity at -16.3 and wash conductivity with elution buffer pH at -14.0. The second of these did not reach significance in the screening design ($p = 0.33$) and reaches it here ($p = 0.005$), which is what the added axial and centre points were there to resolve. No squared term is significant, the smallest p value among the four being 0.14.

Table 5.9: Response-surface coefficients for pool host cell protein.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	141	5.81	24.3	3.21e-12	***
A	52.5	3.96	13.3	6.21e-09	***
B	-49.7	3.96	-12.6	1.21e-08	***
C	4.14	3.96	1.05	0.315	
D	-2.13	3.96	-0.538	0.6	
AB	-16.3	4.2	-3.87	0.00194	**
AC	-4.27	4.2	-1.02	0.328	
AD	-4.33	4.2	-1.03	0.321	
BC	-14	4.2	-3.33	0.00542	**
BD	0.547	4.2	0.13	0.898	
CD	-1.6	4.2	-0.381	0.71	
A ²	-16.4	10.5	-1.56	0.142	
B ²	12.9	10.5	1.24	0.238	
C ²	2.31	10.5	0.221	0.829	
D ²	10.1	10.5	0.968	0.351	

Figure 5.1 shows the fitted surfaces over protein load and elution buffer pH, with wash conductivity and elution stop collect held at their set-points. Pool aggregate rises with both plotted factors and the contours fan out as load increases, which is the interaction between them. Pool host cell protein is flat over elution buffer pH and rises with load, and step yield falls with load and is flat over pH.

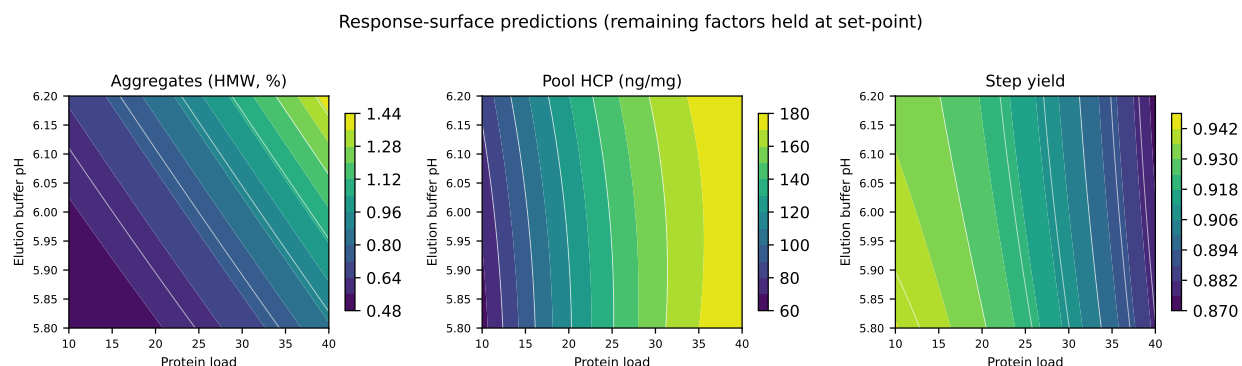


Figure 5.1: Fitted response surfaces over protein load and elution buffer pH, with the remaining factors at their set-points.

Figure 5.2 shows the same three surfaces over protein load and wash conductivity, with elution buffer pH and elution stop collect at their set-points. Pool host cell protein falls steeply from the low conductivity, high load corner to the high conductivity, low load corner, with the spacing of the contours changing with load, which is the interaction. Pool aggregate is unaffected by conductivity, so its contours run horizontally.

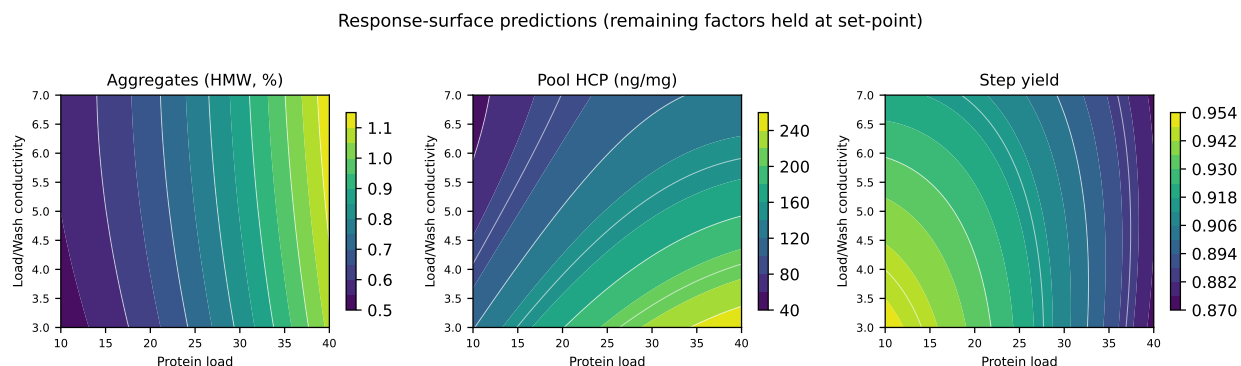


Figure 5.2: Fitted response surfaces over protein load and wash conductivity, with the remaining factors at their set-points.

Figure 5.3 and Figure 5.4 give the residual diagnostics of the two governed responses. The residuals of both models are unstructured against the predicted value, and the normal quantile plots are close to linear. The actual against predicted plots follow the line of identity across the whole range of each response. Neither model shows a heavier residual at high response values, which matters for pool aggregate because the acceptance decision is made at the top of its range.

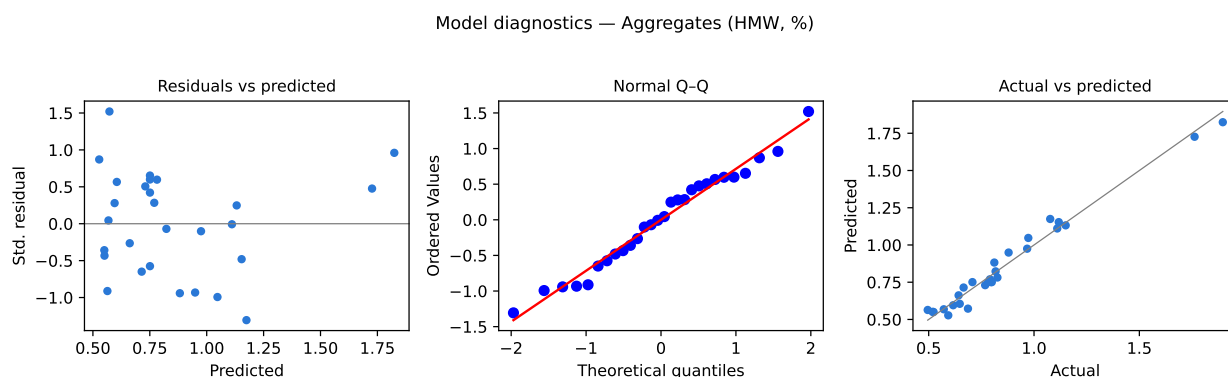


Figure 5.3: Residual diagnostics for the pool aggregate response-surface model.

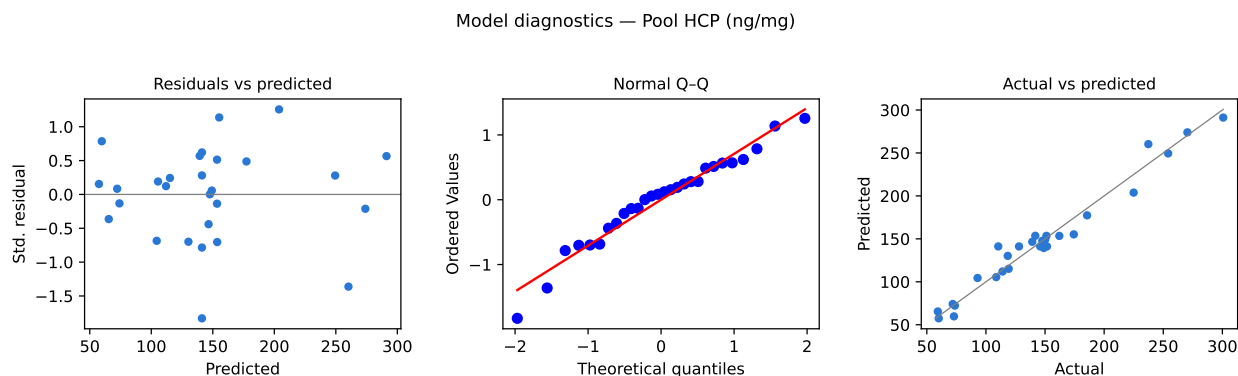


Figure 5.4: Residual diagnostics for the pool host cell protein response-surface model.

At the set-point of every factor the response-surface models predict 0.75 % HMW for pool aggregate, 141 ng/mg for pool host cell protein and 91.7% for step yield. The first two are compared with the in-process criteria in §6 and §7.

5.4 Mechanistic interpretation

The directions of all four effects follow from the physical chemistry of a bind and elute cation exchange separation, and the study confirms them.

Protein load acts on all three responses through one quantity (the fraction of the ligand in use). As the column approaches its capacity the monomer and aggregate peaks broaden and their overlap increases, a larger part of the aggregate front arrives inside the collected pool, and pool aggregate rises. The same broadening loses monomer at both edges of the pool, at the stop collect point and in the fraction cut away with the aggregate, and step yield falls. At high load there is also less free ligand from which the wash can displace the weakly bound host cell protein, more of it co-adsorbs with the antibody, and pool host cell protein rises. Therefore, one physical quantity explains a positive load coefficient on pool aggregate, a positive one on pool host cell protein and a negative one on step yield, which is what Table 5.8 and Table 5.9 show.

Elution buffer pH and elution stop collect act on the aggregate separation and on nothing else. A higher elution pH reduces the net positive charge on the antibody and weakens its binding. Monomer and aggregate then leave the column earlier and closer together, and the resolution on which the separation depends is compressed. A later stop collect takes more of the descending edge of the elution peak into the pool, and the aggregate elutes on that edge. Neither parameter affects the load and wash phase, which is where host cell protein is cleared, and neither has a significant coefficient in the pool host cell protein model.

All three interactions in the pool aggregate model are positive, which follows from the same argument. Each pair of factors acts on the resolution or on the pool cut, and the penalty of raising one is therefore larger when the other has already been raised. Raising the elution pH at a high load moves an already broadened aggregate front

into a pool that is already collecting into the tail. The model puts the interaction of load with elution pH at +0.096 per coded unit, against a main effect of +0.280 for load. The practical consequence is that the aggregate performance of the step cannot be read from any one parameter. It is read from the corner of the operating region, which is why the design space in §6 is stated as a region and not as four independent ranges.

Wash conductivity acts on host cell protein clearance alone, through the ionic strength that shields the charge between the ligand and the weakly bound acidic species. A higher conductivity displaces more of that population during the wash while the antibody, which binds more strongly, stays adsorbed. Its interaction with protein load is negative, which means the benefit of a higher wash conductivity is larger at high load than at low load. That is expected, because a heavily loaded column carries more co-adsorbed host cell protein for the wash to remove. Its interaction with elution buffer pH is also negative, and the mechanism is the same population seen from the other end. Where the wash has already stripped the weakly bound host cell protein, raising the elution pH releases little further, and where it has not, the higher pH carries more of it into the pool.

Residual DNA and leached Protein A behaved as the mechanism predicts. Both are acidic or strongly negatively charged at the operating pH, neither binds the sulfonate ligand, and both were cleared in the load flow through at every condition of both designs. Clearance was $1.5 \log_{10}$ for residual DNA and $1.0 \log_{10}$ for leached Protein A, and neither varied with any studied parameter across the ranges tested. Neither attribute constrains the operating region of this step.

6 Design space

The design space of this step is the region of the four multivariate parameters over which the fitted response-surface models predict a pool that meets both in-process criteria, which are 1 % HMW for aggregate and 193 ng/mg for host cell protein. The region was evaluated on an even grid of 14,641 points over the characterized cube, of which 68.7% meet both criteria. Aggregate is the binding constraint and rejects 21.1% of the cube on its own, while host cell protein rejects 17.0%.

The two constraints bind in different planes, which is why the region is stated as a region. Aggregate is bounded from above by protein load, elution buffer pH and elution stop collect acting together, and Figure 5.1 shows the aggregate surface rising towards the corner where load and elution pH are both high. Host cell protein is bounded from below by wash conductivity and from above by protein load, and Figure 5.2 shows its surface rising towards the corner where load is high and wash conductivity low. Protein load appears in both boundaries, with opposite partners, and is the parameter that carries the operating region.

Three nested regions are used in this report and they are not the same thing. The characterization ranges in Table 4.1 bound the knowledge space, which is the region the study visited. The design space is the part of that region shown above to meet both in-process criteria. The normal operating ranges in Table 4.1 are the ranges the process will be operated within at commercial scale, and lie inside the characterized ranges on every parameter.

The normal operating ranges are not entirely inside the design space, a gap that is reported here and not left to be inferred. Evaluated on the same fitted models over an even grid across the normal operating ranges of the four parameters, 92.8% of that smaller region meets both criteria. The part that does not is the corner at high protein load with a high elution pH and a late stop collect, where the predicted pool aggregate reaches 1.27 % HMW against the in-process limit of 1 %. Aggregate rejects 6.4% of the normal operating region and host cell protein rejects 1.1%, the latter at the opposite corner of low wash conductivity with high load.

Two things bound what that finding means, the first being that the in-process criteria are not the drug substance criteria. The aggregate criterion sits an assurance factor of 5 below the drug substance criterion of 5 % HMW. Therefore, a pool predicted at 1.27 % HMW is above an internal control limit and is not a batch predicted to fail specification. The second bound is that these are mean predictions of a mean-level model, whose reliability at the edge of the shaded region is approximately half if the variation around the mean is symmetric. For this reason the analysis in §7 replaces the mean by a 95 % predictive interval.

The consequence for change control is stated in ICH Q8 (International Council for Harmonisation 2009), where movement within the design space is not a change from a regulatory filing perspective and movement outside it is. Under ICH Q10 (International Council for Harmonisation 2008) any planned movement is still assessed under the pharmaceutical quality system before it is made. Because part of the normal operating range for protein load falls outside the region defined here, the control strategy in §10 holds the routine condition at the set-point and treats the pool aggregate result as the in-process control that confirms the outcome of each batch.

7 Proven acceptable ranges

Table 7.1 gives the proven acceptable range of each response-surface factor against each governed attribute. Two analyses are reported for each pair, of which the first holds the other three parameters at their set-points and scans the parameter of interest across its characterization range on the fitted model. The second varies the other three within their normal operating ranges by Monte-Carlo simulation and requires the 95 % predictive interval of the response to stay inside the criterion. The criterion column is the in-process criterion of §2.2 in both cases, the basis being the same for every row of the table (in-process).

Table 7.1: Proven acceptable ranges by governed attribute and parameter.

CQA	Parameter	Char. range	Unit	Crite- rion	PAR (set-point)	PAR (NOR)
Aggregates (HMW, %)	Protein load	10-40	g/L resin	≤ 1	10-36	10-27.6
Aggregates (HMW, %)	Load/Wash conductivity	3-7	mS/cm	≤ 1	3-7	3-7
Aggregates (HMW, %)	Elution buffer pH	5.8-6.2	pH	≤ 1	5.8-6.2	5.8-6.05
Aggregates (HMW, %)	Elution stop collect	0.5-1.5	OD desc.	≤ 1	0.5-1.5	0.5-1.12
Pool HCP (ng/mg)	Protein load	10-40	g/L resin	≤ 193	10-40	10-28.4
Pool HCP (ng/mg)	Load/Wash conductivity	3-7	mS/cm	≤ 193	3.3-7	4.55-7
Pool HCP (ng/mg)	Elution buffer pH	5.8-6.2	pH	≤ 193	5.8-6.2	5.8-6.2
Pool HCP (ng/mg)	Elution stop collect	0.5-1.5	OD desc.	≤ 193	0.5-1.5	0.537-1.5

Two of the eight rows are bounded inside the characterization range under the first analysis. Protein load is bounded at 36 g/L resin by the aggregate criterion, which is an edge of failure inside a range the process could reach. Wash conductivity is bounded from below at 3.3 mS/cm by the host cell protein criterion. The remaining six rows extend across the whole characterization range at the set-point of the other parameters, which means that no single parameter moved on its own leaves the criterion within the range studied.

By contrast, the second analysis is more demanding, and every range narrows under it. For aggregate, protein load narrows to 10 to 27.6 g/L resin, elution buffer pH to

5.8 to 6.05 and elution stop collect to 0.5 to 1.12 OD on the descending edge. For host cell protein, wash conductivity narrows to 4.55 to 7 mS/cm and protein load to 10 to 28.4 g/L resin.

Comparing those ranges with the normal operating ranges in Table 4.1 gives the same finding as §6 from a different direction. The upper edge of the normal operating range for protein load, at 32 g/L resin, lies above the proven acceptable range for aggregate under the propagated analysis, which ends at 27.6 g/L resin. The lower edge of the normal operating range for wash conductivity, at 4 mS/cm, lies below the proven acceptable range for host cell protein, which starts at 4.55 mS/cm. The upper edges for elution buffer pH and elution stop collect are likewise outside their propagated ranges by a small margin. The set-point of every parameter is comfortably inside every proven acceptable range. However, the propagated analysis does not support the edge of one routine range taken with all three other parameters simultaneously at their own worst positions.

Figure 7.1 shows the aggregate analysis against protein load (the governing parameter for that attribute). The left panel is the analysis at the set-point and the right panel the propagated one, with the acceptable region shaded and the normal operating range marked. The predictive band in the right panel widens the curve enough to move the boundary down by roughly a fifth of the characterization range, which is the size of the penalty that simultaneous variation in the other three parameters carries.

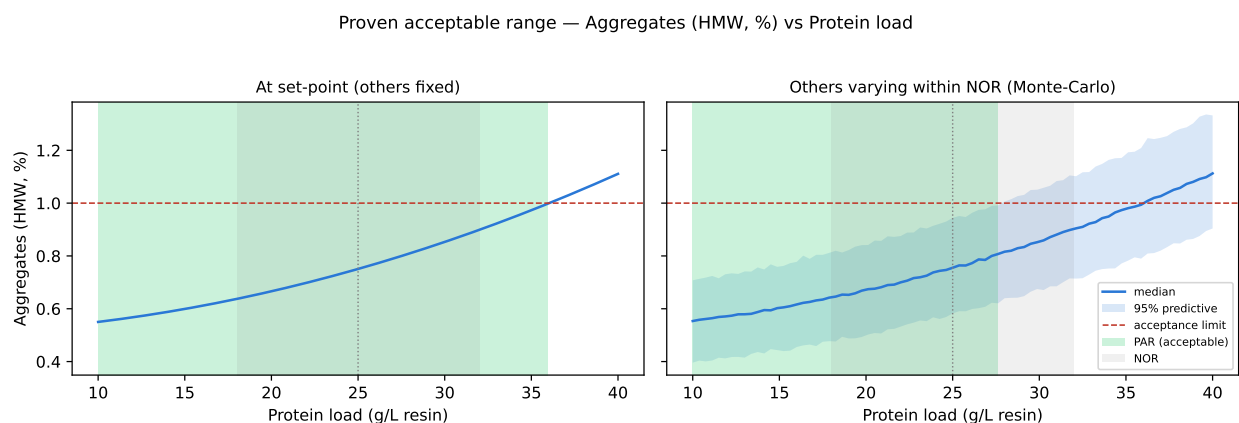


Figure 7.1: Proven acceptable range for pool aggregate against protein load, at set-point (left) and with the other parameters varying within their normal operating ranges (right).

Figure 7.2 shows the host cell protein analysis against wash conductivity, which is its governing parameter. The criterion binds from below rather than from above, because the weakly bound host cell protein is displaced by ionic strength. A conductivity that is too low leaves it on the column to elute with the product.

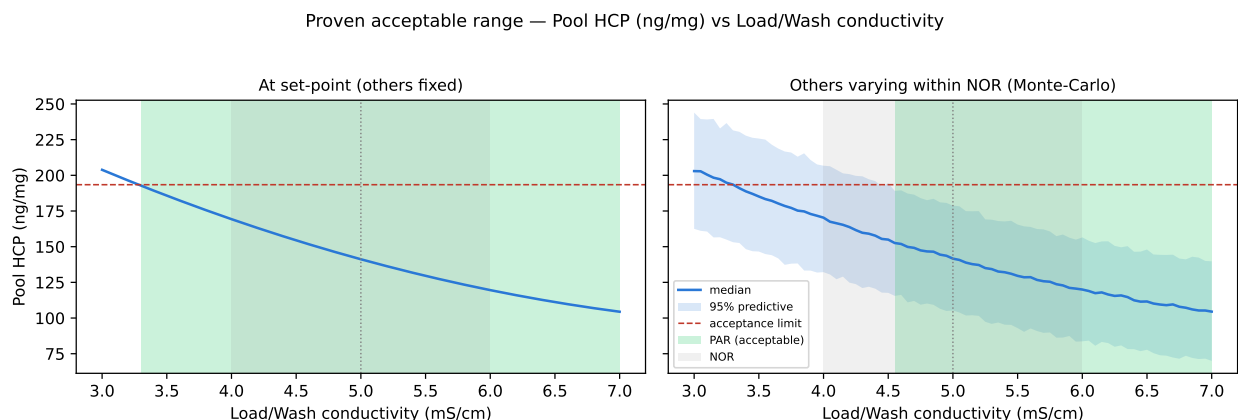


Figure 7.2: Proven acceptable range for pool host cell protein against wash conductivity, at set-point (left) and with the other parameters varying within their normal operating ranges (right).

Figure 7.3 shows the aggregate analysis against elution stop collect, which is the direct handle on the trade between aggregate clearance and yield. Stopping the collection earlier lowers the pool aggregate and loses monomer in the tail, while stopping later does the reverse. The step is operated at 1 OD on the descending edge, which sits inside both analyses.

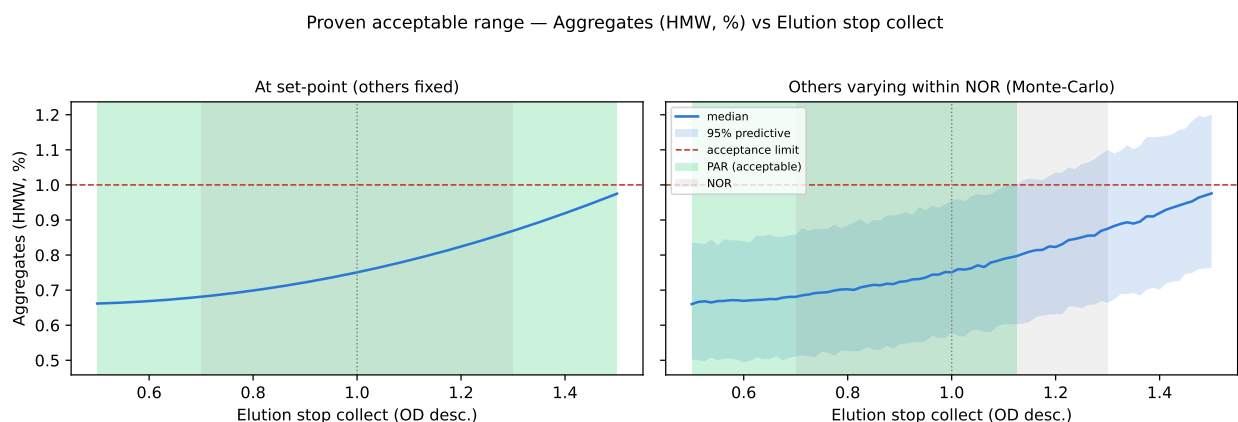


Figure 7.3: Proven acceptable range for pool aggregate against elution stop collect, at set-point (left) and with the other parameters varying within their normal operating ranges (right).

Two limits apply to every range in Table 7.1. The first is that the ranges are univariate statements about one parameter at a time and do not replace the multivariate region of §6, because a combination of parameters each inside its own proven acceptable range can still sit outside the region. The second is that each range stops at the edge of the characterization range, and a row that extends across the whole range is a statement about the range studied and not an unbounded one. Step yield carries no proven acceptable range at all, because it is a process performance attribute with no

quality criterion attached to it. It is reported in §5 only for the effect of protein load on it.

8 Process capability and robustness

Table 8.1 gives the commercial-scale capability of the four attributes this step governs, estimated by simulating 2,000 batches of the whole train with every parameter of every step varying inside its normal operating range. The capability index is one sided in each case, because all four attributes carry an upper limit only.

Table 8.1: Commercial-scale capability of the attributes governed by the step.

CQA	Crit.	Spec	Acceptance	Mean $\pm$ SD	Cpk
Aggregates (HMW)	H	upper	0-5	0.753 ± 0.088	16.1
Host Cell Protein (HCP)	M-H	upper	0-100	15.2 ± 4.6	6.1
Residual DNA	VL	upper	0-0.001	0.0001 ± 0	10.8
Leached Protein A	L-M	upper	0-5	0.003 ± 0.0006	2,785.3

Aggregate has a capability of 16.1 against the drug substance criterion of 5 % HMW, with a simulated mean of 0.75 % and a highest simulated batch of 1.15 %. Host cell protein has a capability of 6.1 and is the tightest of the four, with a simulated mean of 15.2 ng/mg against a criterion of 100 ng/mg. Residual DNA is at 10.8, and leached Protein A at 2785, a figure that should be read as a bound rather than as an estimate, because the simulated mean of 0.003 ppm lies well below the limit of quantitation of the assay (0.25 ppm).

The capability of this step cannot be separated from the capability of the train for three of these four attributes. Host cell protein, residual DNA and leached Protein A are cleared again at anion exchange chromatography, which reduces host cell protein a further 6.2-fold in the nominal process. The figures in Table 8.1 are therefore the outcome of both steps together. Aggregate is the exception, because no step after cation exchange changes it. The aggregate capability of the drug substance is therefore delivered by this step. The only other input to it is the load the step receives from the low-pH inactivation step (Step 6).

The simulation gives the same picture as §6 and §7 at the level of whole batches. All 2,000 simulated batches met the drug substance criterion for all four attributes. However, 0.8% of them carried more aggregate than the in-process limit of 1 % HMW. That is the same small corner of the normal operating range that §6 identified, seen as a batch frequency instead of as a volume of parameter space. The in-process limit is doing what an in-process limit is for, which is to flag those batches for assessment while the drug substance criterion is still met with the margin Table 8.1 reports.

Robustness at the level of a single parameter follows from the same models. Elution flow rate has no detected effect on any response over its characterization range, which is twice the width of its normal operating range, leaving the step robust to it. Wash conductivity has no detected effect on pool aggregate or on step yield, and acts on host cell protein alone. Elution buffer pH and elution stop collect have no detected effect on pool host cell protein or on step yield, and act on aggregate alone. Protein load is the only parameter that acts on all three responses.

The tightest capability of the A-Mab drug substance is 1.51, which belongs to a cumulative viral clearance attribute that this step does not contribute to. That attribute and its contributing steps are reported in PCR-008 and PCR-009 and consolidated in PCMR-001.

9 Parameter classification

Every parameter of the step was classified under SOP-4001 from the effects demonstrated in §5 and from the risk of leaving the operating region during routine manufacture. The classification of each is given in Table 4.1 and the reasoning is below.

- Protein load was classified as a well controlled critical process parameter, because it has the largest effect on pool aggregate, on pool host cell protein and on step yield. It is the parameter that bounds the operating region from above. It is not a critical process parameter, because the load is set by weight from a measured protein concentration on a pooled and assayed intermediate, which leaves the risk of moving outside the range during a batch low.
- Wash conductivity was classified as a well controlled critical process parameter, because it is the governing parameter for pool host cell protein and bounds the operating region from below. It has no effect on pool aggregate or step yield. The buffer is released against a conductivity specification before use and is measured in line during the wash, which leaves the risk of an undetected excursion low.
- Elution buffer pH was classified as a well controlled critical process parameter, because it has the second largest effect on pool aggregate and interacts with protein load. It has no detected effect on the other two responses. Buffer pH is set at preparation, released before use and monitored at the column inlet, which makes it easy to hold inside a range that is narrow in absolute terms.
- Elution stop collect was classified as a well controlled critical process parameter, because it affects pool aggregate directly and is the handle that trades aggregate against yield. It is executed by the chromatography system against an optical density set-point on the descending edge of the peak, which is an automated decision on a continuously measured signal. The parameter is therefore well controlled by construction.
- Elution flow rate was classified as a general process parameter, because no effect on any response was detected over a characterization range twice the width of its normal operating range. The mechanism gives no reason to expect one at this particle size, and it is controlled at its set-point by the pump and monitored as a process performance parameter.

The step therefore carries 4 well controlled critical process parameters, 1 general process parameter and 0 critical process parameters. No parameter of this step was classified as critical, because every quality linked parameter is set on a measured signal or on a released buffer and none of them can move quickly enough during a batch to leave its range undetected. Across the whole characterization campaign the train carries 1 critical process parameter, which is reported in PCMR-001.

10 Contribution to the control strategy

The step contributes five elements to the control strategy for commercial manufacture.

The first element is the set of parameter ranges. The four quality linked parameters are operated within the normal operating ranges of Table 4.1 and are recorded in the batch record. Protein load is the parameter that carries the operating region. However, §6 and §7 both place the upper part of its normal operating range outside the region that the propagated analysis supports. The routine target for load is therefore the set-point and not the upper part of the range. A batch executed above 27.6 g/L resin is assessed against the pool aggregate result before the material is released to the next step.

The second element is in-process monitoring of the two governed responses, in which pool aggregate is measured on every batch by AMV-3011 against an in-process limit of 1 % HMW and pool host cell protein by AMV-3012 against 193 ng/mg. Both limits are derived in §2.2 and both carry an assurance factor below the point at which the drug substance would be at risk. A result above either one triggers an assessment of the batch and not a rejection of it. Aggregate is the more important of the two for this step, because it is the last step that can reduce it.

The third element is the life-cycle control of the column, in which resin lot, packing quality and the number of cycles are controlled under SOP-2008, and column efficiency is verified by plate count and peak asymmetry against the acceptance criteria of that procedure before use. The characterization in this report was executed on a column within those criteria, making the operating ranges valid for a column that meets them. Resin ageing was not a variable of this study and is managed by the resin life-cycle programme rather than by a parameter range.

The fourth element is the control of the material entering the step, which is the neutralized pool of the low-pH viral inactivation step (Step 6). The aggregate content of that pool is an input this step does not control. The nominal load carries 1.91 % HMW and the step reduces it 2.6-fold, which means that an increase in the incoming burden consumes the same margin as an increase in protein load. The in-process limit on the aggregate content of the viral inactivation pool, reported in PCR-006, is the control that bounds this input.

The fifth element is release testing of the drug substance, in which aggregate, host cell protein, residual DNA and leached Protein A are tested against the criteria in Table 2.1. The capability of each is reported in §8, while the attributes this step clears but does not set are also controlled at anion exchange chromatography (PCR-008). The viral safety of the process rests on the low-pH inactivation, anion exchange and

virus filtration steps and not on this one. The contribution of this step to the overall control strategy is consolidated in PCMR-001.

11 Discussion

The step is now characterized over ranges two to three times wider than the ranges it will be operated within, and shows the behaviour the physical chemistry predicts. Four parameters were studied multivariately and all four separate cleanly by mechanism. Protein load acts on every response through the fraction of the ligand in use. Elution buffer pH and elution stop collect act on the aggregate separation and on nothing else, while wash conductivity acts on host cell protein clearance and on nothing else. That separation is what makes the operating region interpretable, because each boundary of it can be attributed to a named physical cause and not only to a fitted coefficient.

Aggregate rather than host cell protein bounds this step, and that has a consequence for the control strategy. Host cell protein is the attribute with the longer history of being difficult at a polishing step. Moreover, it has the tightest capability of the four attributes in Table 8.1, at 6.1. However, it is cleared again downstream, while aggregate is cleared here for the last time. The propagated proven acceptable ranges in §7 narrow faster for aggregate than for host cell protein. A step judged on host cell protein alone would be given a wider operating region than the data supports.

Confidence in the scale-down model rests on three observations, the first being its qualification against at-scale data under SOP-1001 on the input attributes, the output attributes and the performance of the separation. The centre-point means of the response surface design sit close to the values the nominal commercial process delivers, at 0.77 % HMW and 134 ng/mg. The lack of fit test is not significant for any of the three responses, which shows that the run to run spread of the model is large enough to contain the residual of a quadratic surface fitted to it and that no systematic shape was missed.

Three limitations bound the claims made here. The first is that the models are mean-level models, in which the predicted response at a point is a mean and the assurance that a batch at that point meets the criterion is roughly half at the boundary of the region. For this reason the propagated analysis in §7 governs the routine ranges, while the analysis at the set-point is reported beside it and not in place of it. The models are also valid only inside the characterization ranges, because no squared term reached significance and a surface that is planar within a range gives no information about its shape beyond that range. Finally, the design space has not been confirmed at scale at its edges. The commercial process will be executed at the set-point during process performance qualification, while the edges of the region rest on the qualified scale-down model.

Two deviations were recorded and both were retained, with the investigations reported in §13, and neither changed a conclusion. The affected runs were retained in the analysis, and each sits within 0.7 residual standard deviations of the value its model

fits. Neither run is influencing the fitted surfaces in a way that would change an effect estimate or a boundary.

12 Conclusions

The cation exchange polishing step has been characterized on a qualified scale-down model over ranges wider than its intended operating ranges, and meets the five objectives of §1.2. Protein load, elution buffer pH and elution stop collect govern the aggregate content of the pool, wash conductivity and protein load govern the host cell protein content, and protein load governs the step yield. Elution flow rate has no detected effect on any of them.

The design space is the region of the four parameters over which both in-process criteria are met, and covers 68.7% of the characterized region, with aggregate as the binding constraint. Proven acceptable ranges were established for each parameter against each governed attribute in two analyses. The propagated analysis places the upper part of the normal operating range for protein load, elution buffer pH and elution stop collect, and the lower part of the range for wash conductivity, outside the supported region. The routine condition is the set-point, which sits inside every range by a clear margin.

At commercial scale all 2,000 simulated batches meet the drug substance criteria for the four attributes this step governs, with a capability of 16.1 for aggregate and 6.1 for host cell protein, the tightest of the four. 4 parameters were classified as well controlled critical process parameters and 1 as a general process parameter, with no critical process parameter at this step. The 2 recorded deviations were investigated and retained, leaving the outcome of this step to be carried into the master report PCMR-001 in support of Stage 2.

13 Deviations from the plan

2 deviations from PCP-007 were recorded during execution and are listed in Table 13.1. Both were detected after the affected run had been executed, both were investigated under the site deviation procedure, and both were dispositioned as retained, which means the affected run stayed in the reported analysis. No deviation invalidated a design and no study was re-executed.

Table 13.1: Deviations recorded during execution.

Devia- tion	Summary	Detected during	Disposi- tion
DEV-007-01	Expired Tris load/wash buffer lot used in screening run 6	post-execution documentation review	retained
DEV-007-02	Column inlet temperature excursion during RSM run 14	equipment history review	retained

13.1 DEV-007-01, expired load and wash buffer lot

Screening run 6 was executed with a lot of Tris load and wash buffer, LOT-BUF-2287, that had passed its expiry date of 2026-02-11 before the run was performed. The lot was not identified at preparation and the deviation was found in the documentation review after the design had been executed. The root cause is that the expiry date was not verified when the buffer was issued to the run, which is a control failure in the issuing step and not a defect of the buffer itself.

The investigation followed SOP-2103, which authorises re-test of a retained sample when an expiry deviation is raised. The retained sample of LOT-BUF-2287 was re-tested for the two attributes that determine how the buffer behaves in this step. It returned a pH of 5.02 and a conductivity of 5.06 mS/cm, both within the release specification the lot had met when it was prepared, which shows that the buffer had not changed in the properties the step depends on.

The impact on the study was assessed from the fitted screening model, because run 6 is a factorial corner run that carries weight in every effect estimate. Its measured responses sit within 0.7 residual standard deviations of the values the screening model fits for that condition, across all three responses. A run that the model reproduces to within its own residual spread is not distorting the effect estimates. The direction and the significance of every screening effect in §5.2 are unchanged whether the run is treated as conforming or not. The run was retained, with the deviation closed by a corrective action on the buffer issuing step under SOP-2103.

13.2 DEV-007-02, column inlet temperature excursion

During response surface run 14 the column inlet temperature rose to 26.4 °C and stayed above the controlled range for 38 minutes, which covers the load and part of the wash phase of that run. The excursion was found in the equipment history review of EQ-CHR-118 (the temperature-controlled chromatography chamber) and not during the run. The root cause was a drift of the chamber controller. The chamber itself was within its calibration at execution, with calibration next due on 2026-05-30, which shows that the drift was in the controller set-point and not in the measurement.

Temperature affects a cation exchange separation through the viscosity of the mobile phase and through the binding equilibrium, with a higher temperature sharpening the peaks slightly and weakening binding slightly. Both would act on the aggregate result of the affected run, which is why the impact was assessed on that response. Three verification runs were executed at the condition of run 14 with the chamber under control. The pools were assayed by AMV-3221, a verification qualified size exclusion method with a precision of 3.8 % and a limit of quantitation of 0.20 % HMW (Appendix D). On 3 runs that precision gives a 95 % confidence half-width on the mean of 9.44 % relative, which is the resolution with which the verification can detect a shift.

No shift larger than that resolution was found between the affected run and the verification runs. The measured responses of run 14 also sit within 0.5 residual standard deviations of the values the response surface model fits for that condition, across all three responses. The run is consistent with the surface fitted to the other 27 runs of the design. The run was retained, the controller was recalibrated and the chamber set-point was added to the pre-run verification checklist for the step under SOP-2010.

The two deviations are independent. One is a material control failure detected in documentation review and the other an equipment control failure detected in equipment history review. They affect different designs and different runs, and neither shares a root cause with the other. Neither affects the design space of §6, the proven acceptable ranges of §7 or the classification of §9.

Appendices

Appendix A — Screening design matrix and responses

Table 13.2 gives the coded settings of the 19 screening runs, with the factor letters of Table 4.2, and Table 13.3 gives the measured responses of the same runs.

Table 13.2: Screening design, coded factor settings.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

Table 13.3: Screening design, measured responses.

Run	Aggregates (HMW, %)	Pool HCP (ng/mg)	Step yield
1	0.5	136	0.936
2	0.604	114	0.962
3	0.548	135	0.94
4	0.818	138	0.973
5	0.452	66.2	0.934
6	0.564	61.3	0.951

Run	Aggregates (HMW, %)	Pool HCP (ng/mg)	Step yield
7	0.579	48.8	0.946
8	0.752	67.6	0.934
9	0.792	428	0.883
10	1.1	247	0.888
11	1.14	301	0.923
12	1.79	240	0.908
13	0.807	105	0.895
14	1.04	113	0.893
15	1.12	115	0.884
16	1.83	149	0.886
17	0.833	153	0.929
18	0.829	183	0.935
19	0.692	129	0.91

Appendix B — Response-surface design matrix and responses

Table 13.4 gives the coded settings of the 28 response surface runs and Table 13.5 the measured responses of the same runs.

Table 13.4: Response-surface design, coded factor settings.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0

Run	Type	A	B	C	D
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

Table 13.5: Response-surface design, measured responses.

Run	Aggregates (HMW, %)	Pool HCP (ng/mg)	Step yield
1	0.518	109	0.98
2	0.593	114	0.955
3	0.572	151	0.973
4	0.818	142	0.942
5	0.495	59.3	0.921
6	0.616	71.9	0.946
7	0.686	60.1	0.932
8	0.812	73	0.925
9	0.768	237	0.87
10	0.973	254	0.878
11	1.15	301	0.9
12	1.76	270	0.89
13	0.826	174	0.902
14	1.12	139	0.903
15	1.08	118	0.891
16	1.9	119	0.844
17	0.523	73.7	0.935
18	1.11	186	0.879
19	0.666	225	0.92
20	0.791	92.9	0.909
21	0.647	149	0.921
22	0.879	148	0.914
23	0.642	162	0.935
24	0.967	150	0.922
25	0.796	146	0.881
26	0.708	110	0.952
27	0.782	128	0.921
28	0.8	152	0.917

Appendix C — Full effect and coefficient tables

Table 13.6, Table 13.7 and Table 13.8 give every estimated screening effect for the three responses, in order of absolute size. Table 13.9 gives the response-surface coefficients for step yield, which is the response not tabulated in §5.3.

Table 13.6: Screening effects, pool aggregate, all terms.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	0.599	0.299	0.024	12.5	1.6e-06	***
C	0.34	0.17	0.024	7.09	0.000103	***
D	0.319	0.16	0.024	6.65	0.000161	***
AC	0.196	0.0979	0.024	4.08	0.00354	**
AD	0.154	0.0772	0.024	3.22	0.0123	*
CD	0.131	0.0657	0.024	2.74	0.0256	*
B	-0.0185	-0.00924	0.024	-0.385	0.71	
BC	0.0152	0.00758	0.024	0.316	0.76	
BD	-0.0131	-0.00657	0.024	-0.274	0.791	
AB	0.012	0.00602	0.024	0.251	0.808	

Table 13.7: Screening effects, pool host cell protein, all terms.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-127	-63.3	8.85	-7.15	9.7e-05	***
A	116	58.2	8.85	6.58	0.000173	***
AB	-56.7	-28.4	8.85	-3.21	0.0125	*
BD	39.6	19.8	8.85	2.24	0.0557	
D	-25.8	-12.9	8.85	-1.46	0.182	
AD	-24.6	-12.3	8.85	-1.39	0.202	
CD	24.4	12.2	8.85	1.38	0.205	
BC	18.4	9.18	8.85	1.04	0.33	
AC	-12.5	-6.25	8.85	-0.707	0.5	
C	-9.45	-4.72	8.85	-0.534	0.608	

Table 13.8: Screening effects, step yield, all terms.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	-0.0518	-0.0259	0.00287	-9.04	1.79e-05	***
BC	-0.0121	-0.00607	0.00287	-2.12	0.0672	
B	-0.0111	-0.00553	0.00287	-1.93	0.0899	
AD	-0.00914	-0.00457	0.00287	-1.59	0.15	
D	0.00683	0.00342	0.00287	1.19	0.268	
C	0.00666	0.00333	0.00287	1.16	0.279	
BD	-0.00566	-0.00283	0.00287	-0.988	0.352	

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
CD	-0.00473	-0.00236	0.00287	-0.824	0.434	
AC	0.00401	0.002	0.00287	0.699	0.504	
AB	3.38e-05	1.69e-05	0.00287	0.00589	0.995	

Table 13.9: Response-surface coefficients for step yield.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.917	0.00651	141	4.36e-22	***
A	-0.0307	0.00444	-6.92	1.05e-05	***
B	-0.00757	0.00444	-1.71	0.112	
C	-0.00358	0.00444	-0.807	0.434	
D	-0.00548	0.00444	-1.24	0.239	
AB	0.00803	0.00471	1.71	0.112	
AC	0.000269	0.00471	0.0571	0.955	
AD	-0.000697	0.00471	-0.148	0.885	
BC	-0.00646	0.00471	-1.37	0.193	
BD	0.00185	0.00471	0.393	0.701	
CD	-0.00658	0.00471	-1.4	0.185	
A ²	-0.0103	0.0117	-0.882	0.394	
B ²	-0.00278	0.0117	-0.237	0.816	
C ²	0.000575	0.0117	0.049	0.962	
D ²	0.0108	0.0117	0.924	0.372	

Table 13.10: Analysis of variance and lack-of-fit test for step yield.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	0.0217	14	0.00155	4.38	0.00573
Residual	0.00461	13	0.000355	nan	nan
Lack of fit	0.0021	10	0.00021	0.251	0.958
Pure error	0.00251	3	0.000836	nan	nan

Appendix D — Analytical methods summary

Table 13.11 gives the validated method used for each analyte of the step and the form in which each result is reported.

Table 13.11: Analytical methods, analytes and reportable results.

Reference	Validated method	Analyte	Reportable result
AMV-3011	Size-Variants (SEC-HPLC)	Size variants (monomer, dimer, higher aggregate)	Pool aggregate, % HMW
AMV-3012	Host-Cell Protein ELISA	Host cell protein	Pool host cell protein, ng/mg
AMV-3014	Residual DNA (qPCR)	Residual host cell DNA	Pool residual DNA, ng per dose
AMV-3016	Leached Protein A by ELISA (ppm)	Leached Protein A	Pool leached Protein A, ppm

Method performance is quoted in this report where the campaign method register carries it. The host cell protein immunoassay AMV-3012 has a precision of 9.5 % and a limit of quantitation of 1.0 ng/mg. The leached Protein A immunoassay AMV-3016 has a precision of 6.5 % and a limit of quantitation of 0.25 ppm. The verification qualified size exclusion method AMV-3221, used only in the investigation of DEV-007-02 in §13.2, has a precision of 3.8 % and a limit of quantitation of 0.20 % HMW.

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